

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: COMPUTER NETWORKS

Course Credits: 3

Semester: III

Course Category: Core

Type of Course: Lecture / Practical

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	3
Tutorial	0
Practical	2
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	40%
Entrepreneurship Skills	30%
Skill Development	30%

Course Relevance

Level	Percentage Relevance
Global	40%
National	30%
Regional	20%
Local	10%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
C01	Describe the basics of data communication, networking, internet and their importance	BL-2
C02	Explain the concepts of layered architecture, protocols and interworking in computer networks	BL-2
C03	Examine the various networks using the logical addressing by applying subnetting and routing concepts	BL-4
C04	Demonstrate the working of transport and application layer protocols in an IP based networking infrastructure.	BL-3
C05	Assess application layer services, client-server model, HTTP, email, WWW, TELNET.	BL-5

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: Data communications	characteristics, components, data representation, data flow	Activity: Online Certification Coursera – 13 to 15 Hours	BL-2	CO1	NIL
	Networks : distributed processing,		BL-2	CO1	
	network criteria, types of connections,		BL-4	CO1	
	types of topologies,		BL-4	CO3	
	categories of networks		BL-3	CO3	
	Network models : the OSI model		BL-3	CO4	

	layered architecture		BL-4	CO4	
	layers in the OSI model		BL-4	CO4	
	TCP/IP protocol suite.		BL-4	CO5	
Module 2: Physical layer	analog and digital, analog signals, digital signals, analog versus digital, data rate limit	Assessment : Internal Test Activity: Flip Class/ Mini project/ Presentation / Seminar	BL-4	CO1	NIL
	transmission impairments, transmission mode		BL-4	CO2	
	modulation of digital data,		BL-3	CO2	
	telephone modems		BL-4	CO2	
	modulation of analog signal		BL-4	CO2	
	FDM, WDM, TDM,		BL-4	CO2	
	guided media, unguided media		BL-2	CO1	
	switching		BL-2	CO1	
	networking devices.		BL-2	CO1	

Module 3: Data link layer	error detection and correction	Activity: Presentation / Report writing, Mini project/ Case studies.	BL-3	CO4	NIL
	types of errors		BL-4	CO4	
	error detection and correction techniques		BL-3	CO4	
	data link control and protocols		BL-3	CO4	
	flow and error control		BL-4	CO4	
	stop-and-wait ARQ		BL-4	CO4	
	go-back-n ARQ		BL-4	CO4	
	selective repeat ARQ		BL-4	CO4	
	multiple access		BL-4	CO4	
Module 4: Network layer	classful addressing, logical addressing, IPv4, subnets	Activity: Presentation / Report writing, Mini project/ Case studies.	BL-2	CO1	NIL
	FLSM, VLSM, classless inter domain routing (CIDR)		BL-3	CO3	
	public and private addresses, network address translation (NAT)		BL-3	CO3	
	unicast routing protocols, distance vector routing, RIP, link state routing		BL-3	CO3	
	OSPF, path vector routing, BGP		BL-3	CO3	

	Transport layer : process-to-process delivery, port addresses		BL-4	CO4	
	socket address, user datagram protocol (UDP), transmission control protocol (TCP)		BL-3	CO3	
	3-way handshaking-SCTP-data traffic, traffic descriptors-		BL-2	CO1	
	congestion control		BL-2	CO1	
Module 5: Application layer	domain name system (DNS)	Assessment : Preparatory Exam Activity: Presentation / Report writing, Mini project/ Case studies.	BL-3	CO1	NIL
	Dynamic Host Configuration Protocol (DHCP)		BL-3	CO1	
	remote logging, TELNET		BL-3	CO1	
	electronic mail, file transfer		BL-3	CO1	
	WWW, HTTP, HTTPS		BL-4	CO1	
	Network security : Basic of cryptography		BL-2	CO1	
	types of Cryptography		BL-4	CO2	
	symmetric key cryptography		BL-3	CO2	
	public key cryptography		BL-4	CO2	

Textbooks:	
1	Behrouz A.Forouzan, "Data Communications and Networking",McGrawHill,5TH Edition,2017, ISBN-10: 1259064751.
2	Rick Graziani, Allan Johnson,"Routing Protocols and Concepts CCNA Exploration Companion Guide", Pearson, 2008,ISBN-13: 978-1-58713-204-9.

CO-PO/PSO Mapping

Course Outcome s	Blooms Taxonom y Level	Programme Outcomes (PO)								
		P O 01	P O 02	P O 03	P O 04	P O 05	P O 06	P O 07	P O 08	P O 09
CO1	L2	3	1	1	1	1	1	-	-	-
CO2	L3	2	2	2	1	1	2	-	-	-
CO3	L4	2	2	2	1	2	2	2	1	2
CO4	L5	3	3	3	2	3	2	3	1	2
CO5	L5	3	3	2	2	2	2	2	1	2